

INNOVA JUNIOR COLLEGE
 JC 2 PRELIMINARY EXAMINATION 2
 in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

CLASS

INDEX NUMBER

CHEMISTRY

9647/01

Paper 1 Multiple Choice

18 September 2013

1 hour

Additional Materials: Data Booklet
 Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
 Write in soft pencil.
 Do not use staples, paper clips, highlighters, glue or correction fluid.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.
 Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
 Any rough working should be done in this booklet.

WORKED SOLUTIONS

This document consists of **18** printed pages.



Innova Junior College

[Turn over

Section A

For each question there are four possible answers, **A**, **B**, **C**, and **D**. Choose the **one** you consider to be correct.

- 1 A giant molecule contains a large amount of carbon, mainly of isotopes ^{12}C and ^{13}C . It was found that the relative atomic mass of carbon in the molecule is 12.20.

What is the ratio of ^{12}C and ^{13}C ?

- A** 3:1 **B** 3:4 **C** 4:1 **D** 1:4

Answer: **C**

Let the relative abundance of ^{12}C be x and ^{13}C be $1-x$

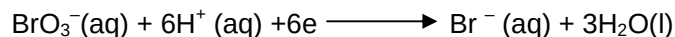
$$12x + 13(1-x) = 12.20$$

$$12x + 13 - 13x = 12.20$$

$$x = 0.80$$

$$\begin{aligned} \text{Ratio of } ^{12}\text{C} : ^{13}\text{C} &= 0.80 : (1-0.80) \\ &= 0.80 : 0.20 \\ &= 4 : 1 \end{aligned}$$

- 2 20.0 cm^3 of 0.02 mol dm^{-3} aqueous sodium bromate(V), NaBrO_3 , was found to react completely with 80.0 cm^3 of 0.01 mol dm^{-3} hydroxylamine, NH_2OH . The half equation for the reduction of bromate(V) ions is given as shown.



Which of the following could be the nitrogen containing product in this reaction?

- A** N_2O
B NO_2
C NO
D NO_3^-

Answer: **C**

$$\begin{aligned} \text{Amount of } \text{BrO}_3^- \text{ used} &= 0.02/1000 \times 20 \\ &= 0.0004 \text{ mol} \end{aligned}$$

$$\text{Amount of electrons taken in} = 0.0004 \times 6 = 0.0024 \text{ mol}$$

$$\begin{aligned} \text{Amount of } \text{NH}_2\text{OH} \text{ used} &= 0.01/1000 \times 80.0 \\ &= 0.0008 \text{ mol} \end{aligned}$$

$$\begin{aligned} \text{Amount of } \text{NH}_2\text{OH} : \text{Amount of electrons} \\ 0.0008 : 0.0024 \\ 1 : 3 \end{aligned}$$

→ 3 electrons were given out by each N in NH_2OH

Oxidation number of N in $\text{NH}_2\text{OH} = -1$

New oxidation number of N = +2

- 3 The successive ionisation energies (IE) of two elements, **A** and **B**, are given below.

IE/ kJ mol ⁻¹	1st	2nd	3rd	4th	5th	6th	7th	8th
A	1090	2350	4610	6220	37800	47000	-	-
B	1251	2298	3822	5158	6542	9362	11018	33604

What is the likely formula of the compound that is formed when **A** reacts with **B**?

- A** **AB** **B** **A₂B₃** **C** **AB₄** **D** **A₄B**

Answer: **C**

Biggest Jump for Q: 4th to 5th IE: Q is in group IV

Biggest Jump for R: 7th to 8th IE: R is in Group VII

- 4 Which electronic configuration represents an atom of an element that forms a simple ion with a charge of -3?

- A** 1s² 2s² 2p⁶ 3s² 3p¹
B 1s² 2s² 2p⁶ 3s² 3p³
C 1s² 2s² 2p⁶ 3s² 3p⁶ 3d¹ 4s²
D 1s² 2s² 2p⁶ 3s² 3p⁶ 3d³ 4s²

Answer: **B**

Simple ion means that it will not be a transition metal; hence **C** and **D** will be eliminated.

In order to form a charge of -3, it will gain 3 electrons to fulfil octet configuration, hence it has 5 valence electrons.

- 5 Which of the following species has a square planar structure?

- A** BrF₄⁻
B BF₄⁻
C CH₄
D C₂H₄

Answer: **A**

BrF₄⁻ has 2 lone pair and 4 bond pair, hence square planar structure.

BF₄⁻ and CH₄ has 4 bond pair → tetrahedral

C₂H₄ has 3 bond pair around each C → square planar wrt to each C

- 6 Which of the following processes is endothermic?
- A condensation of steam
- B freezing of water
- C** electrolysis of water
- D $\text{Ca(s)} + 2\text{H}_2\text{O(l)} \longrightarrow \text{Ca(OH)}_2\text{(aq)} + \text{H}_2\text{(g)}$

Answer: C

The electrolysis of water involves $2\text{H}_2\text{O(l)} \rightarrow 2\text{H}_2\text{(g)} + \text{O}_2\text{(g)}$ which requires electrical energy, thus endothermic.

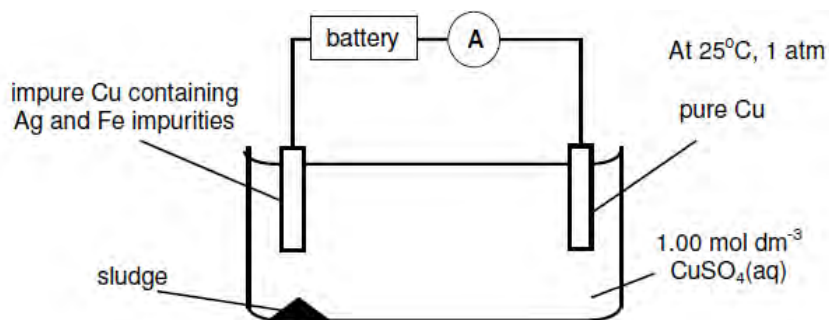
- 7 Which value would be required to estimate the lattice energy for the hypothetical ionic compound MgH_2 ?
- A** The electron affinity of hydrogen
- B The first ionisation energy of hydrogen
- C The magnesium-hydrogen bond energy
- D The standard enthalpy change of formation of MgH

Answer: A

the hypothetical ionic compound MgH consists of Mg^{2+} and H^- ions. Thus, to estimate the LE of MgH_2 , we require the value of electron affinity of hydrogen.

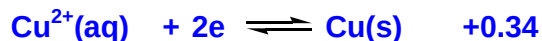
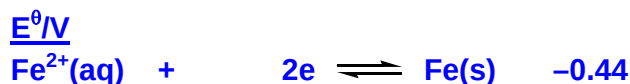
- 8 Use of the Data Booklet is relevant to this question.

The circuit shown in the diagram was set up.



What are the products formed at the anode and cathode and what is the substance present in the sludge after some time?

	anode	cathode	sludge
A	Cu^{2+}	Cu	Ag
B	Fe^{2+}	Cu^{2+}	Ag
C	$\text{Ag}^+, \text{Fe}^{2+}$	Cu^{2+}	Zn
D	$\text{Cu}^{2+}, \text{Fe}^{2+}$	Cu	Ag



At the anode:



Since the $E^0_{\text{Fe}^{2+}/\text{Fe}}$ is more negative than $E^0_{\text{Cu}^{2+}/\text{Cu}}$, Fe is oxidised to Fe^{2+} ions at the anode as well.

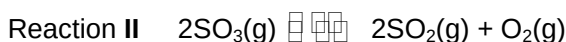
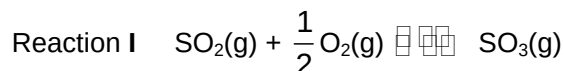
Since the $E^0_{\text{Ag}^+/\text{Ag}}$ is more positive than $E^0_{\text{Cu}^{2+}/\text{Cu}}$, Ag will not be oxidised to Ag^+ ions at the anode but will drop just off at the anode as "anode sludge".

At the cathode:



Since $E^0_{\text{Cu}^{2+}/\text{Cu}}$ is more positive than $E^0_{\text{Fe}^{2+}/\text{Fe}}$, Fe^{2+} will remain in the solution as Cu^{2+} is preferentially reduced over Fe^{2+} .

- 9 The equilibrium constant for reaction I below is K . What is the equilibrium constant for reaction II?



A K^{-2}

C $2K$

B K^2

D $2K^{-1}$

Kc for Reaction I:

$$K_c = \frac{[\text{SO}_3]}{[\text{SO}_2][\text{O}_2]^{1/2}} = K \quad \text{----- (1)}$$

Kc for Reaction II:

$$K_c = \frac{[\text{SO}_2]^2 [\text{O}_2]}{[\text{SO}_3]^2}$$

Rearranging (1):

$$\frac{[\text{SO}_2] [\text{O}_2]^{1/2}}{[\text{SO}_3]} = \frac{1}{K}$$

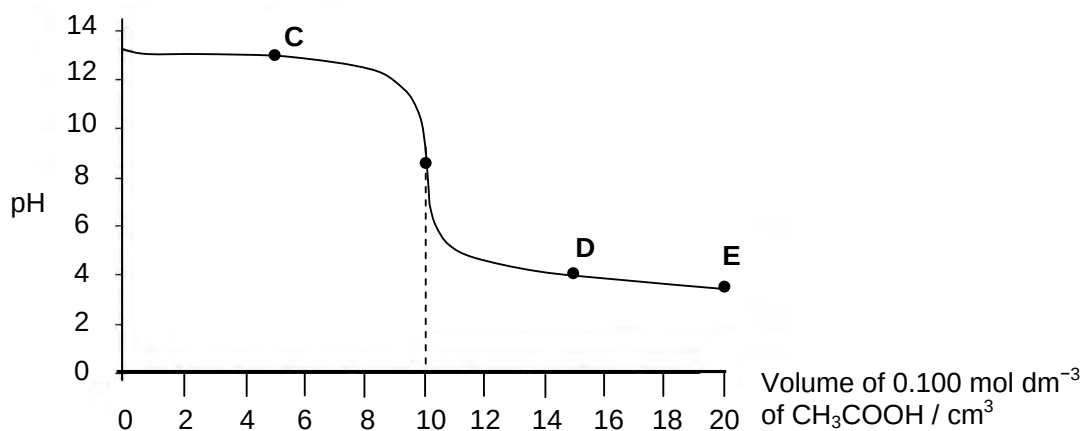
Squaring on both sides:

$$\frac{[\text{SO}_2]^2 [\text{O}_2]}{[\text{SO}_3]^2} = \frac{1}{K^2} = K^{-2} \quad \text{which is the Kc}$$

expression for Reaction II

Answer : A

- 10 When 5.00 cm³ of a 0.100 mol dm⁻³ base is titrated against 0.100 mol dm⁻³ of aqueous CH₃COOH, the following titration curve is obtained.



Which of the following correctly identifies the base used and the point on the curve at which pH = pK_a?

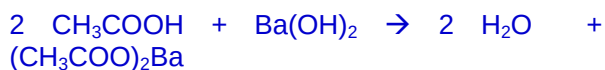
	base used	point where pH = pK _a
A	KOH(aq)	C
B	Ba(OH) ₂ (aq)	D
C	KOH(aq)	E
D	Ba(OH) ₂ (aq)	E

Amt of base in conical flask = $(5/1000) \times 0.1 = 5 \times 10^{-4}$

Amt of CH₃COOH needed for complete neutralisation = $(10/1000) \times 0.1 = 1 \times 10^{-3}$

Base : CH₃COOH = $5 \times 10^{-4} : 1 \times 10^{-3} = 1 : 2$

Thus, the base is **dibasic**.



Amt of (CH₃COO)₂Ba formed at neutralisation = 5×10^{-4}

Amt of CH₃COO⁻ = $2 \times 5 \times 10^{-4} = 1 \times 10^{-3}$

For pH=pK_a, [CH₃COOH] = [CH₃COO⁻].

At point D:

Amt of excess CH₃COOH = $(5/1000) \times 0.1 = 5 \times 10^{-4}$

Amt of CH₃COOH ≠ Amt of CH₃COO⁻

⇒ [CH₃COOH] ≠ [CH₃COO⁻]

⇒ pH ≠ pK_a

At point E:

Amt of excess CH₃COOH = $(10/1000) \times 0.1 = 1 \times 10^{-3}$

Amt of CH₃COOH = Amt of CH₃COO⁻

⇒ [CH₃COOH] = [CH₃COO⁻]

⇒ pH = pK_a

Answer : D

- 11 The table gives the concentrations and pH values of the aqueous solutions of two compounds, **C** and **D**. Either compound could be an acid or a base.

	C	D
Concentration	2 mol dm ⁻³	2 mol dm ⁻³
pH	6	9

Student **E** concluded that **C** is a strong acid.

Student **F** concluded that the extent of dissociation is lower in **C(aq)** than in **D(aq)**.

Which of the students is/are correct?

- A** **E** only
B **F** only
C Both **E** and **F**
D Neither **E** nor **F**

From the pH values, C is an acid while D is a base. To determine degree of dissociation in C: pH = 6 $\Rightarrow [H^+] = 10^{-6}$ Since $[H^+] \ll [C]$, compound C is not a strong acid. Thus, student E is wrong.	To determine degree of dissociation in D: pH = 9 $\Rightarrow pOH = 14 - 9 = 5$ $\Rightarrow [OH^-] = 10^{-5}$ Since $[OH^-] \ll [D]$, compound D is not a strong base. The degree of dissociation of C lesser than D. Thus, student F is correct. Answer : B
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- 12 A 10 cm³ sample of 0.010 mol dm⁻³ HCl is diluted by adding distilled water at constant temperature.

Which one of the following items correctly shows the effect of the dilution on the concentrations of H⁺ and OH⁻ ions in the solution?

	[H ⁺]	[OH ⁻]
A	increase	increase
B	increase	decrease
C	decrease	increase
D	decrease	decrease

Answer: C

HCl is a strong acid & so it's completely ionized in aqueous solution.

On dilution, [H⁺] decreases.

Since temperature is constant, $K_w = [H^+][OH^-]$ does not change.

Hence as [H⁺] decreases, [OH⁻] must increase to keep K_w constant.

- 13** Lead is the final product formed by a series of changes in which the rate determining stage is the radioactive decay of uranium-238. This radioactive decay is a first order reaction with a half-life of 4.5×10^9 years.

What would be the age of a rock sample, originally lead-free, in which the molar proportion of uranium to lead is now 1:3?

- A** 1.5×10^9 years
B 2.25×10^9 years
C 4.5×10^9 years
D 9.0×10^9 years

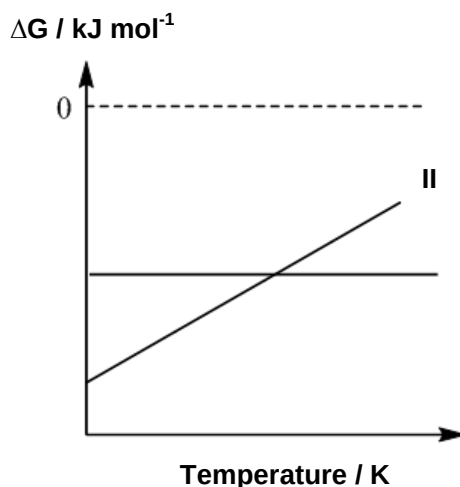
Answer: D

Uranium	lead
1	0
$\frac{1}{2}$	$\frac{1}{2}$
$\frac{1}{4}$	$\frac{1}{2} + \frac{1}{4} = \frac{3}{4}$

For molar ratio of Ur: Pb to be 1:3, Ur needs to undergo 2 half-lives of decay.

Thus, age of rock sample = $2(4.5 \times 10^9) = \underline{9.0 \times 10^9 \text{ years}}$

- 14** Ellingham diagrams are plots of ΔG against temperature, T. An Ellingham diagram for two reactions involving the oxidation of S and SO_2 is as shown below:



Given that $\Delta G = \Delta H - T\Delta S$, which of the following correctly shows the two reactions corresponding to reactions I and II in the above Ellingham diagram?

	reaction I	reaction II
A	$2\text{S(s)} + 3\text{O}_2\text{(g)} \longrightarrow 2\text{SO}_3\text{(g)}$	$2\text{SO}_2\text{(g)} + \text{O}_2\text{(g)} \longrightarrow 2\text{SO}_3\text{(g)}$
B	$2\text{SO}_2\text{(s)} + \text{O}_2\text{(g)} \longrightarrow 2\text{SO}_3\text{(g)}$	$\text{S(s)} + \text{O}_2\text{(g)} \longrightarrow \text{SO}_2\text{(g)}$
C	$\text{S(s)} + \text{O}_2\text{(g)} \longrightarrow \text{SO}_2\text{(g)}$	$2\text{S(s)} + 3\text{O}_2\text{(g)} \longrightarrow 2\text{SO}_3\text{(g)}$
D	$2\text{SO}_2\text{(g)} + \text{O}_2\text{(g)} \longrightarrow 2\text{SO}_3\text{(g)}$	$2\text{S(s)} + 3\text{O}_2\text{(g)} \longrightarrow 2\text{SO}_3\text{(g)}$

Answer: C

Using $\Delta G = \Delta H - T\Delta S$, and the graph of ΔG against T

$$y = mx + c \rightarrow y = \Delta G, c = \Delta H, x = T, m = -\Delta S$$

$\Rightarrow -\Delta S$ is the gradient So just focus on ΔS

For **graph I**, there's no change in entropy.

$\Rightarrow$ no. of gas molecules at reactant side = no. of gas molecules at product side $\Rightarrow$ answer is **C**

For **graph II**, there is a decrease in entropy (ΔS negative) for $2S(s) + 3O_2(g) \rightarrow 2SO_3(g)$,
so $-\Delta S$ has a positive slope gradient.

- 15** When one mole sample of hydrated chromium(III) chloride, $CrCl_3 \cdot 6H_2O$ is treated with excess of aqueous silver nitrate, 3 moles of silver chloride is precipitated. What is the most probable formula of the complex ion in the salt?

- A** $[CrCl_2(H_2O)_4]^+$
- B** $[CrCl(H_2O)_5]^{2+}$
- C** $[CrCl(H_2O)_3]^{2+}$
- D** $[Cr(H_2O)_6]^{3+}$

Answer : D

Keyword: 3 moles of AgCl precipitated means that 3 moles of Cl^- is ionic bonded to the complex
(ie. 3 Cl^- ions are free ions)

From the formula of $CrCl_3 \cdot 6H_2O$, the 3 Cl^- is ionic bonded to the complex & water is dative covalent bonded to the complex.

Thus, answer is $[Cr(H_2O)_6]^{3+}$. The structural formula of the salt is $[Cr(H_2O)_6]^{3+} \cdot 3Cl^-$

- 16** The properties of the oxides of four elements **K**, **L**, **M** and **N** in the third period in the Periodic Table are given below.

- The oxide of **K** is insoluble in water and in dilute acid but is soluble in hot and concentrated sodium hydroxide.
- The oxide of **L** is amphoteric.
- The oxide of **M** reacts with dilute sodium hydroxide at room temperature.
- The oxide of **N** dissolves in water to form a strong alkaline solution.

Which of the following is correct in order of increasing proton number?

- A** **N**, **L**, **K**, **M**
- B** **K**, **L**, **M**, **N**
- C** **N**, **K**, **L**, **M**
- D** **N**, **M**, **K**, **L**

Answer: A

This is a recall question.

Oxide of K insoluble in water and in dilute acid => giant covalent structure

Oxide of K reacts with dilute hot and concentrated sodium hydroxide => it's SiO_2

Oxides of L is amphoteric => Al_2O_3

Oxide of M reacts with dilute sodium hydroxide at room temperature => covalent structure with acidic nature => M is non-metal

The oxide of N dissolves in water to form a strong alkaline solution => ionic structure => N is metal

So arrange according to increasing proton number: metal (ie N), Al (ie L), Si (ie K), non-metal (ie M)

17 The size of Na^+ , Mg^{2+} and Al^{3+} is in the order: $\text{Na}^+ > \text{Mg}^{2+} > \text{Al}^{3+}$. Which of the following best explain this trend?

- A** The number of electrons and protons decreases.
- B** The number of electrons and protons increases.
- C** The number of electrons decreases while the number of protons increases.
- D** The number of electrons are the same but the number of protons increases.

Answer: D

For ionic radius, always must check if they're isoelectronic series

Na^+ , Mg^{2+} and Al^{3+} have 10 electrons but their proton number increases, thus nuclear attraction increases from Na^+ , Mg^{2+} and Al^{3+} . And size is in the order: $\text{Na}^+ > \text{Mg}^{2+} > \text{Al}^{3+}$.

18 Which property of Group II elements (magnesium to barium) or their compounds increases with increasing proton number?

- A** the magnitude of the enthalpy change of hydration of the metal ion
- B** the stability of the carbonate to heat
- C** the acidity of an aqueous solution of the chloride
- D** the tendency to form complex ions

Answer: B

Option A wrong because magnitude of ΔH_{hyd} of M^{2+} ($\propto$ charge/ ionic radius) decrease down group II

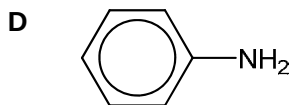
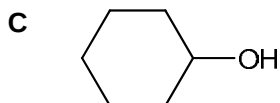
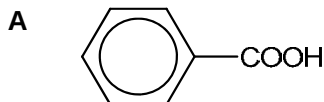
Option B correct because down group II, charge density & polarising power of M^{2+} decrease. So more heat energy required to decompose the group II carbonates. In other words, group II carbonates less easily decompose & more stable towards heat.

→ so lower tendency to form complex ion down group II.

- D 8**

, minimum 7 carbons.

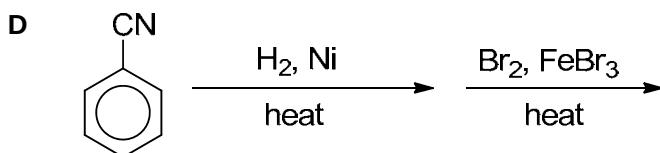
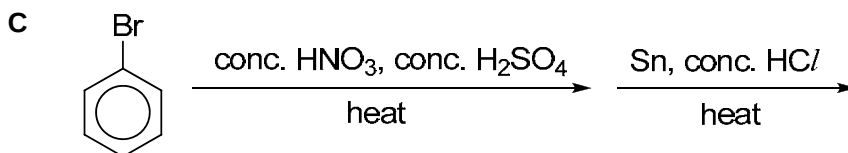
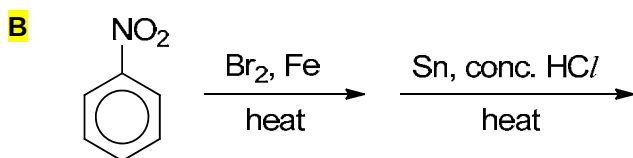
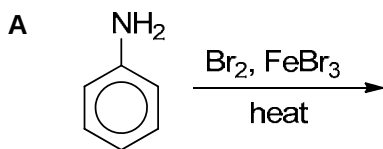
- 21 Compound **O** dissolves in aqueous sodium hydroxide to form a homogenous solution. When benzoyl chloride is added to the resulting solution, a white suspension forms. Which of the following could be compound **O**?



Answer: B

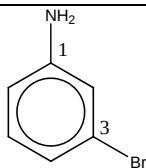
Compound **O** dissolves in aqueous sodium hydroxide => it is acidic, so option C & D out
Only phenol can react with benzoyl chloride (an acyl chloride) to form ester.

- 22 Which of the following reaction schemes produces 3-bromophenylamine in the highest yield?

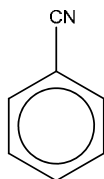


Answer: B

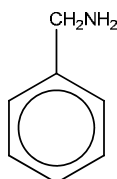
3-bromophenylamine looks like this:



To put bromine to the 3-position, the starting reactant must be electron-withdrawing group (with multiple bond/unstaturated), so eliminate option A & C.



gives



Option D is out because reduction of

which is not phenylamine.

- 23** High energy radiation in the stratosphere produces free-radicals from chlorofluoroalkanes, commonly known as CFCs.

Which free radical is most likely to result from the irradiation of $\text{CHFClCF}_2\text{Cl}$?

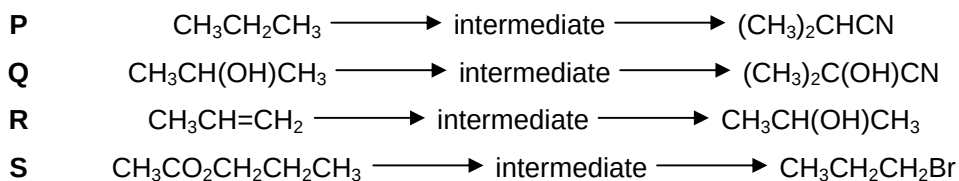
- A $\text{CHFCl}\dot{\text{C}}\text{FCl}$
 B $\dot{\text{C}}\text{HClCF}_2\text{Cl}$
C $\dot{\text{C}}\text{HFCF}_2\text{Cl}$
 D $\dot{\text{C}}\text{FCF}_2\text{Cl}$

Answer: C

To form free radical, you need to undergo homolytic fission, ie breaking of weakest covalent bond.

Check Data Booklet, C-Cl bond is the weakest & will break first to form Cl radical & $\bullet\text{CHFClCF}_2\text{Cl}$

- 24** Which pair of reactions could have the same common intermediate?



A P and Q

C Q and S

B P and R

D R and S

Answer: B

Reaction S is definitely out because ester functional group can only undergo hydrolysis to give RCOOH (CH_3COOH) & $\text{R}'\text{OH}$ ($\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ – the alcohol intermediate) which react with HBr to give $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$.

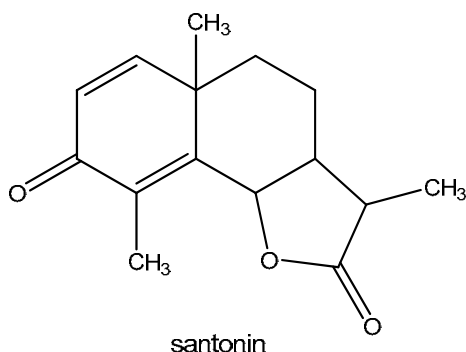
Reaction P, Q, R has intermediate that is alkyl halide functional group.

Reaction P intermediate is $(\text{CH}_3)_2\text{CHBr}$

Reaction Q intermediate is $(\text{CH}_3)_2\text{C}(\text{OH})\text{Br}$

Reaction R intermediate is $\text{CH}_3\text{CH}(\text{Br})\text{CH}_3$ which is also the same as $(\text{CH}_3)_2\text{CHBr}$.

25 Santonin is a drug that was once widely used to expel parasitic worms from the body.



When santonin is first treated with warm dilute H_2SO_4 , and then the product of this reaction is treated with cold acidified KMnO_4 , a final product **T** is obtained.

How many atoms of hydrogen in each molecule of product **T** can be displaced with sodium metal?

A 2

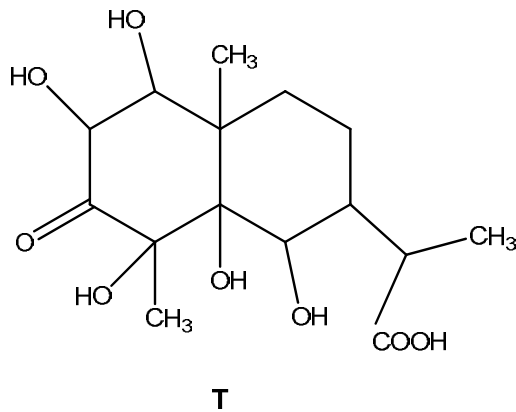
C 5

B 4

D 6

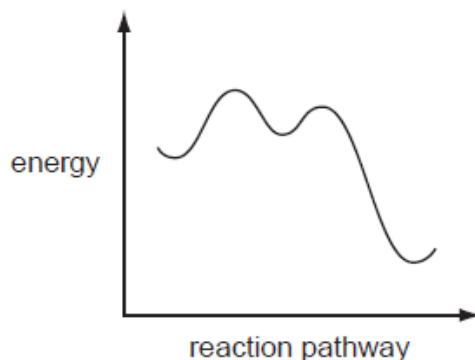
Ans: D

Santonin under acidic hydrolysis (of ester) to give an alcohol and carboxylic acid. The product undergoes mild oxidation (of alkene) with cold acidified KMnO_4 to give **T**.

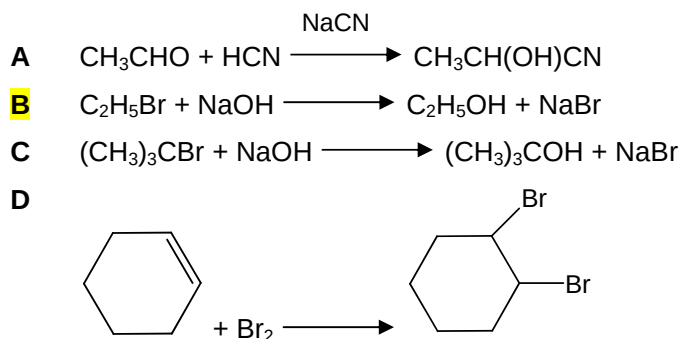


Na undergoes redox with all $-\text{OH}$ groups. Since there are 5 alcohol and 1 carboxylic acid group, a total of 6 H atoms can be displaced with Na .

26 A reaction pathway diagram is shown.



Which reaction does **not** have such a profile?



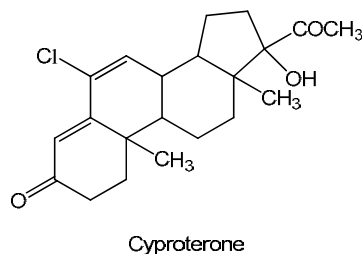
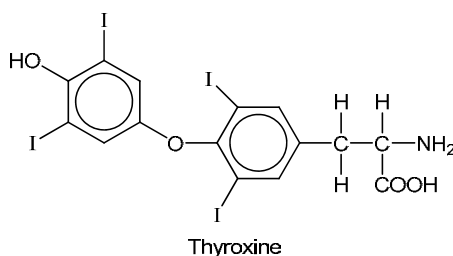
Ans: B

The energy profile diagram suggested that the reaction mechanism is a **two steps reaction** (2 humps).

Option A (Nucleophilic addition), Option C (Nucleophilic substitution, SN1) and Option D (Electrophilic addition) are all two-steps reaction.

Option B (Nucleophilic substitution, SN2) is a single step reaction, hence there should only be one hump in the energy profile diagram.

27 Thyroxine and Cyproterone are compounds that are often used in hormone therapy.



Which of the following reagents **cannot** be used to distinguish between the two compounds?

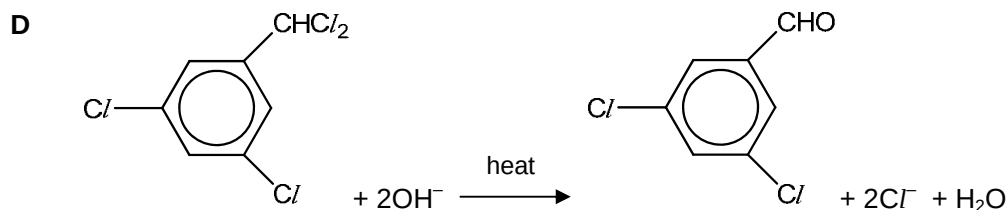
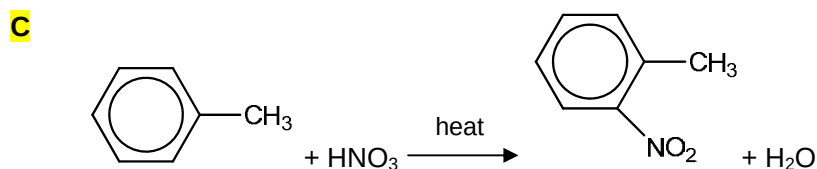
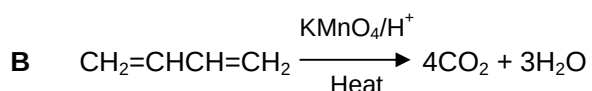
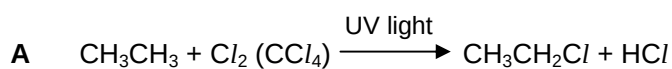
- A** neutral FeCl_3 **C** aqueous silver nitrate
B alkaline aqueous iodine **D** sodium carbonate

Ans: C

Reagents	Thyroxine	Cyproterone
neutral FeCl_3 (test for phenol)	Violet colouration	No violet colouration
alkaline aqueous iodine $\begin{array}{c} \text{OH} \\ \\ \text{H}_3\text{C}-\text{C}-\text{R/H} \\ \\ \text{H} \end{array} \quad \text{or} \quad \begin{array}{c} \text{O} \\ \\ \text{H}_3\text{C}-\text{C}-\text{R/H} \end{array}$ (test for)	No pale yellow ppt	Pale yellow ppt
aqueous silver nitrate (test for acyl chloride / halogenoalkanes)	No (yellow) ppt	No (white) ppt. p orbital of C/I overlap with p orbital of adjacent C atom, strengthening C–C/I bond
sodium carbonate (test for carboxylic acid)	Effervescence.	No effervescence

28 The reaction conditions for four different transformations are given below.

Which transformation has a set of conditions that is **not** correct?



Ans: C

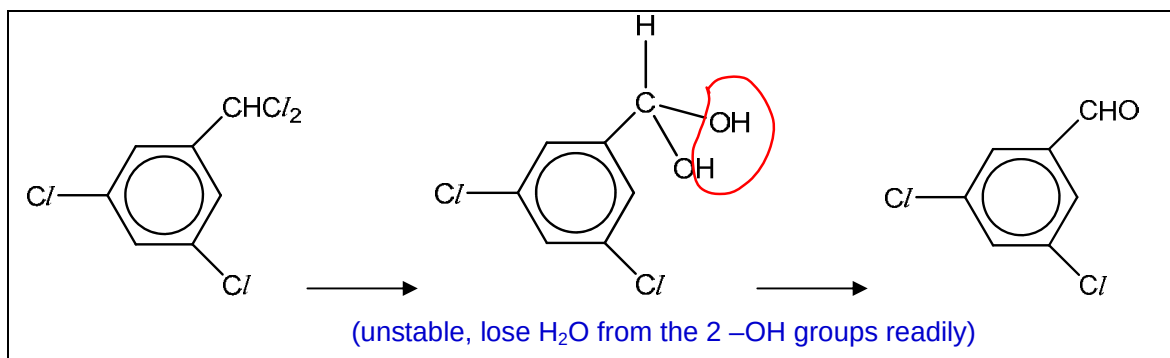
Option A: Free radical substitution. UV light and CCl_4 as solvent are both correct.

Option B: Oxidative cleavage. Acidified KMnO_4 and heating are both correct.

Option C: Electrophilic substitution. **Concentrated** HNO_3 and **concentrated** H_2SO_4 (as a catalyst) at **30°C** is the correct set of reagents and conditions.

Option D: Nucleophilic substitution. Aqueous OH^- and heating are both correct.

Note: Geminal diol (with 2 –OH groups attached to the same C atoms are unstable, and **will undergo internal dehydration readily to give a carbonyl compound**)



- 29 The C–H bond lengths of four hydrocarbons are given in the table below.

compound	structural formula	C–H bond length /nm
methane	CH_4	0.110
ethane	CH_3CH_3	0.110
ethene	$\text{CH}_2=\text{CH}_2$	0.108
ethyne	$\text{CH}\equiv\text{CH}$	0.106

Which of the following helps to explain the shortest C–H bond length observed in ethyne?

- A** Ethyne is a linear molecule.
- B** The orbital of the carbon atom used in formation of the C–H bond in ethyne has the greatest s-orbital character.
- C** An $\text{sp}^2\text{-sp}^2$ overlap is observed between the two carbon atoms in ethyne.
- D** The carbon-carbon triple bond in ethyne is the strongest.

Ans: B

Option A: True statement but does not account for the observation.

Option B:

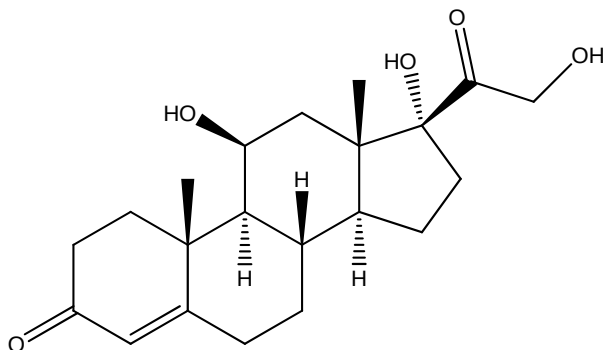
compound	structural formula	Orbitals to form C–H bond	% of s character in hybrid orbital of C
methane	CH_4	sp^3 of C and 1s of H	25%
ethane	CH_3CH_3	sp^3 of C and 1s of H	25%
ethene	$\text{CH}_2=\text{CH}_2$	sp^2 of C and 1s of H	33%
ethyne	$\text{CH}\equiv\text{CH}$	sp of C and 1s of H	50%

The greater the s character in the hybrid orbital of C atom, the shorter the C–H bond will be. (as electrons in the bond will be closer to the nucleus)

Option C: It should be an sp-sp overlap between the two C atoms in ethyne.

Option D: True statement but does not account for the observation.

- 30 Hydrocortisone is a steroid hormone produced by the adrenal gland and is released in response to stress. It is commonly used as an active ingredient in anti-inflammatory creams.



Hydrocortisone

Which of the following statements about hydrocortisone is **incorrect**?

- A It turns hot acidified $K_2Cr_2O_7$ from orange to green.
- B When treated with $NaBH_4$ in the presence of methanol, it forms a compound containing five hydroxy groups.
- C The number of chiral centres increases by 2 after it reacts with cold HCN , in the presence of $NaCN$.
- D** When treated with an excess of hot concentrated acidified $KMnO_4$, it forms a compound containing five carbonyl groups.

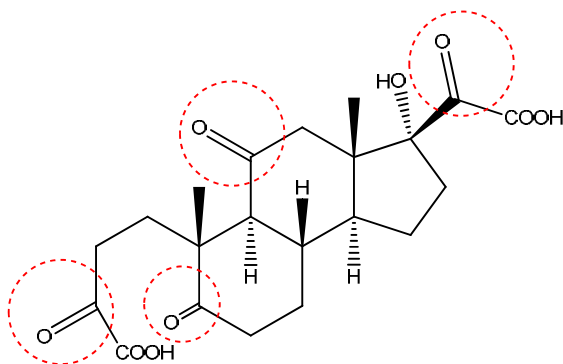
Ans: D

Option A: The primary and secondary alcohols in hydrocortisone can undergo oxidation with hot acidified $K_2Cr_2O_7$, hence turning it from orange to green.

Option B: $NaBH_4$ is a specific reducing agent that only reduces carbonyl groups. The 2 ketone groups in hydrocortisone can be reduced to 2 secondary alcohol groups. Thus in total, there are now 5 $-OH$ groups.

Option C: The 2 ketone groups in hydrocortisone undergoes nucleophilic addition with cold HCN , $NaCN$ to give 2 cyanohydrin groups, each of which contains a chiral C. Thus, number of chiral C in the compound increases by 2.

Option D: Hydrocortisone undergoes oxidation (of alkene and alcohol) to give the following compound containing 4 carbonyl groups.



- 31 Boron is a non-metallic element which is placed above aluminium in Group III of the Periodic Table. It forms a compound with nitrogen known as boron nitride which has a graphite structure.

Which of the following conclusions can be drawn from this information?

- 1 The empirical formula of boron nitride is BN.
- 2 The boron and nitrogen atoms are likely to be arranged alternatively in a hexagonal pattern.
- 3 Boron nitride has a layer structure with van der Waals' forces between the layers.

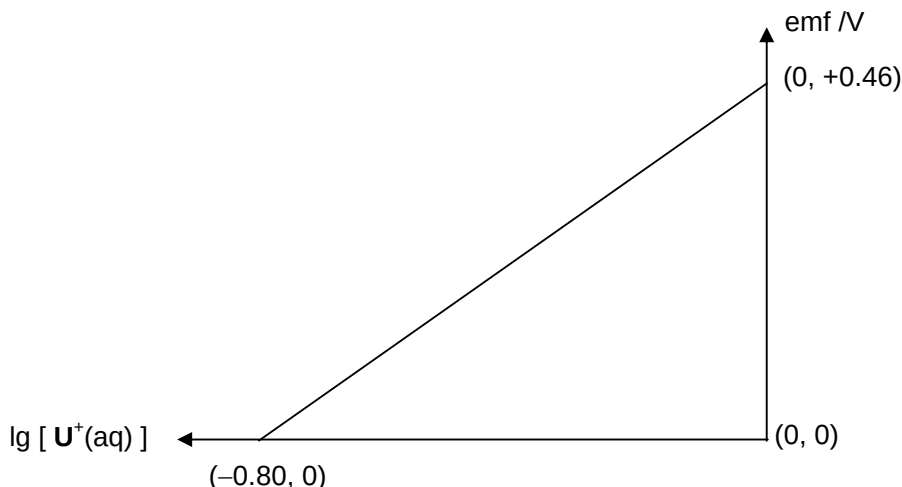
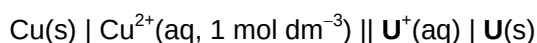
Answer: A

In graphite, the carbon atoms are arranged in hexagonal flat parallel layers. Within each layer, each C atom is covalently bonded to 3 other C atoms. The adjacent layers are held together by weak van der Waals' forces.

Since boron nitride is described to have a graphite structure, all the above statements are correct.

- 32 Use of the Data Booklet is relevant to this question.

The graph below shows the variation in electromotive force (emf) of the electrochemical cell with $\lg [\text{U}^+(\text{aq})]$ at 298 K.



Which of the following statements are true?

- 1 Cu(s) is the negative electrode.
- 2 The emf of the given cell under standard conditions will be +0.46V.
- 3 The standard electrode potential of the $\text{U}^+(\text{aq}) \mid \text{U(s)}$ half-cell is +0.80V.

Ans: A

Option A: From cell notation, Cu^{2+}/Cu half-cell is the **oxidation half-cell** of the **galvanic cell** (since it's being placed on the LHS of the cell notation). Thus Cu(s), being the **anode**, is the negative electrode.

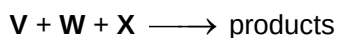
Option B: Under standard conditions, $[U^+] = 1 \text{ mol dm}^{-3}$. Thus $\lg [U^+(\text{aq})] = 0$. From the graph, when $\lg [U^+(\text{aq})] = 0$, the emf of the cell is +0.46 V.

Option C: Since $E^\ominus_{\text{cell}} = +0.46\text{V}$ under standard conditions,

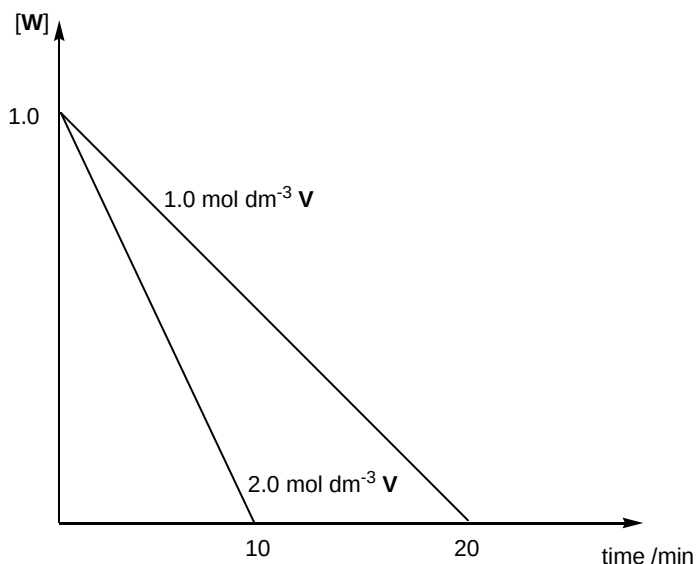
$$+0.46 = E^\ominus(U^+/U) - (+0.34)$$

$$E^\ominus(U^+/U) = +0.80 \text{ V}$$

33 Consider the reaction



Two separate experiments were carried out using 1.0 mol dm^{-3} and 2.0 mol dm^{-3} of **V** respectively. A graph of the variation of the concentration of **W** with time is plotted.



From the graph, which of the following is/are **true** regarding the above reaction?

- 1 The reaction is first order with respect to **V**.
- 2 The reaction is zero order with respect to **W**.
- 3 The reaction is first order with respect to **X**.

Ans: **B**

Option 1: When $[V]$ doubles from 1 mol dm^{-3} to 2 mol dm^{-3} , keeping $[W]$ constant, the rate doubles (gradient doubles) from $\frac{1}{20}$ to $\frac{1}{10}$. Hence reaction is first order wrt **V**.

Option 2: Graph of $[W]$ vs time is a straight line with constant gradient. Hence zero order wrt **W**.

Option 3: There is not sufficient information from the question to determine order of reaction wrt **X**.

34 Which of the following are typical properties of transition elements?

- 1 They exhibit variable oxidation states in their compounds.
- 2 All ions formed by the elements have partially filled d-orbitals.
- 3 Their atoms have fully filled d-orbitals.

Ans: D

Option 1: Transition elements exhibit variable oxidation states in their cpds due to the similarity in energy levels of 3d and 4s electrons. These electrons can be lost easily to form variable oxidation states.

Option 2: The word "All" is wrong. Transition metal ions that are colourless have the electronic configuration of $3d^0$ (empty) or $3d^{10}$ (fully filled).

Option 3: Not all transition element atoms have fully filled d-orbitals. In general, they have the **valence** electronic configuration of $nd^x (n+1)s^y$ (e.g. $3d^x 4s^y$), where $x = 1$ to 10 , $y = 1$ or 2

35 Which of the following regarding the chlorides of the third period elements are **correct**?

- 1 All the chlorides undergo hydrolysis in water except sodium chloride.
- 2 $AlCl_3$ exists only in the gas phase at high temperature.
- 3 The covalent character increases.

Ans: A

Option 1: NaCl undergo hydration in water to form a neutral solution with pH 7. The charge density of the Na^+ ion is too low to undergo hydrolysis.

Option 2: Aluminium chloride, $AlCl_3$ exists as a dimers, Al_2Cl_6 in the vapour state held together by dative bonds at temperatures around 180 - 190°C. As the temperature increases, the dimers, Al_2Cl_6 dissociates to form $AlCl_3$ molecules. Hence Option 2 is correct.

Option 3: Across a period (Na to S), the elements change from metals to non-metals. Bonding of their oxides varies from **ionic to covalent (acidic to basic)**.

36 Which one of the following statements is/are likely to be true with regards to astatine or its compounds?

- 1 Astatine is a black solid at room conditions.
- 2 Hydrogen astatide decompose into its constituent elements readily.
- 3 Sodium astatide reacts with hot concentrated sulfuric acid to produce astatine.

Ans: A

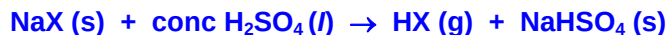
This is a recall question!

Option 1: Colour intensity of halogens increases down the group. Since iodine is already a black solid, astatine must also be a black solid at rtp.

Option 2: Down the group, thermal stability of the HX decreases, as the H-X bond length increases, H-X bond strength decreases, less energy required to break the H-X bond. Hence HAt should decompose into H_2 and At_2 easily.

Option 3:

When concentrated sulfuric acid is added to solid halides, the following reaction occurs.

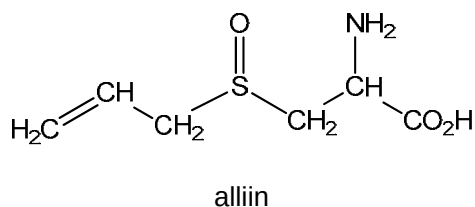


HX produced will continue to reduce H_2SO_4 , depending on the reducing power of the X^- ion.

As the reducing power of X^- increases down the group, following the trend, Astatine will be formed when sodium astatide reacts with hot concentrated sulfuric acid.



37 Alliin is the compound responsible for the smell of garlic.

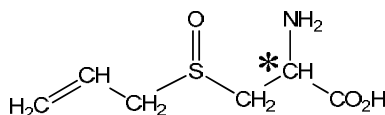


Which statements about this compound are **correct**?

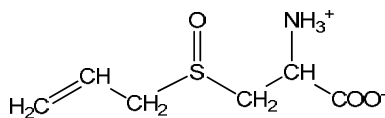
- 1** It can exist in optically active forms.
- 2** It exists as a dipolar ion in aqueous solutions.
- 3** The bond angle $\text{O} - \text{S} - \text{C}$ is 120° .

Ans: **B**

Option 1: There is a chiral centre present, hence it can exist as enantiomers, hence it can exist in optically active forms.

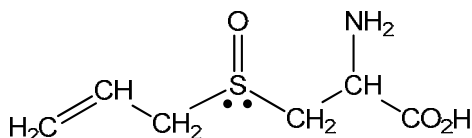


Option 2: There is a NH_2 and COOH group in the same molecule. Hence, intramolecular acid-base reaction can happen to give a zwitterion. A zwitterion is a dipolar ion (has both positive and negative charge).

**Option 3:**

S atom has 1 more lone pair of electrons, but is not drawn in the diagram. Do note that the qn will NOT draw it for you. You will need to realise that on your own.

With a lone pair, S atom has 3 bond pairs, and 1 lone pair $\rightarrow$ shape is trigonal pyramidal (bond angle about 107°)



38 The reaction between carbonyl compounds and 2,4-dinitrophenylhydrazine is known as condensation. Which of the following are intermediate steps found in the multi-step mechanism of this reaction?

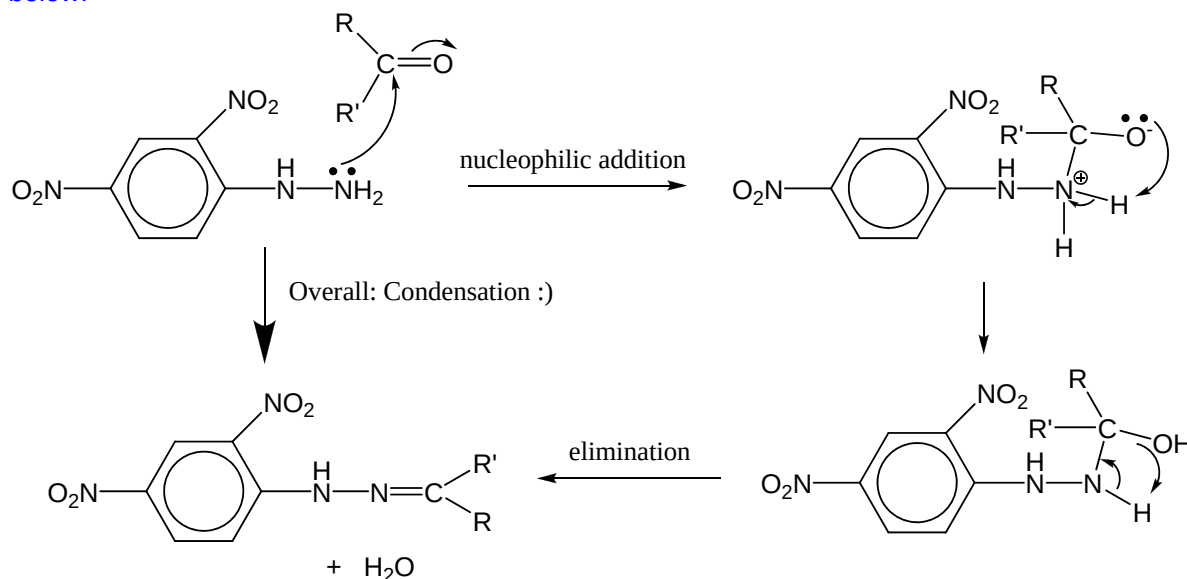
- 1 nucleophilic substitution
- 2 nucleophilic addition
- 3 elimination

Ans: C

Thinking process: Carbonyl compounds undergo nucleophilic addition reaction.

Hence, option 1 is definitely wrong.

For reaction between carbonyl cpds and 2,4-dinitrophenylhydrazine, it follows the mechanism below:



39 Human serum albumin, HSA, is an important protein that transports long chain fatty acids and other hydrophobic molecules in the bloodstream.

Five of the most common amino acids in HSA molecule are listed below.

amino acid	formula of side chain, R in RCH(NH ₂)CO ₂ H
glutamic acid	-CH ₂ CH ₂ CO ₂ H
leucine	-CH ₂ CH(CH ₃) ₂
lysine	-CH ₂ CH ₂ CH ₂ CH ₂ NH ₂
valine	-CH(CH ₃) ₂
cysteine	-CH ₂ SH

Which of the following statements about the type of interactions present in HSA is correct?

- 1 There exists hydrogen bonding between lysine and glutamic acid.
- 2 There are van der Waals' forces between leucine and valine in the protein.
- 3 There are disulfide linkages present in HSA.

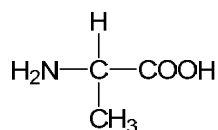
Ans: C

Option 1: $-\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{NH}_2$ in lysine and $-\text{CH}_2\text{CH}_2\text{CO}_2\text{H}$ in glutamic acid will undergo **ACID-BASE reaction** first to give $-\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{NH}_3^+$ in lysine and $-\text{CH}_2\text{CH}_2\text{CO}_2^-$ in glutamic acid. They will subsequently bond to each other via **ionic bonding**.

Option 2: Leucine and valine R groups are non-polar, hence they will interact with each other via van der Waals' forces of interaction.

Option 3: There is cysteine amino acid present, hence there will be disulfide linkages present.

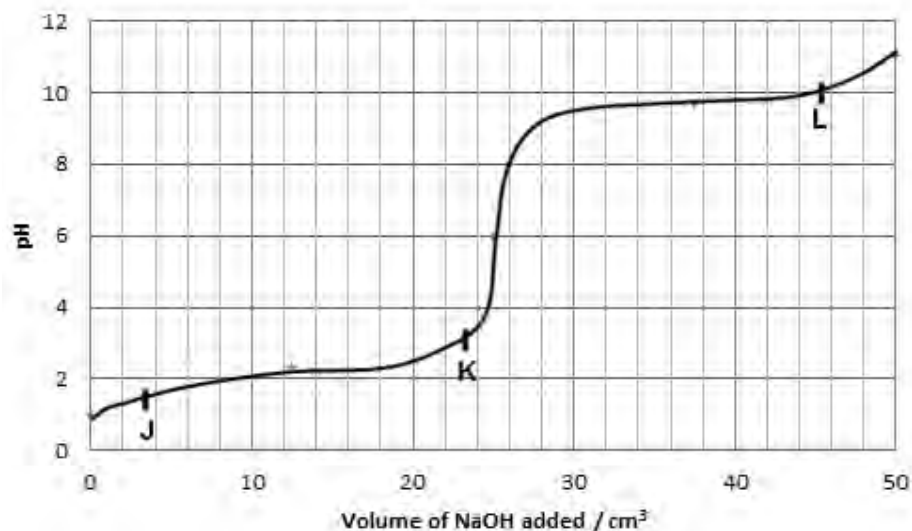
- 40 Alanine is a non-essential amino acid which is found in a wide variety of foods, especially meats.



alanine

A fully protonated alanine has 2 pKa values associated with it: 2.3 and 9.7.

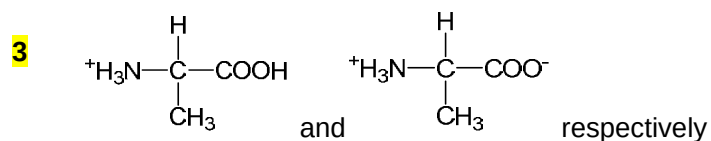
The pH change when 1 mol dm^{-3} of NaOH is added to 50 cm^3 of a 1 mol dm^{-3} solution of alanine in a solution of pH 1 is shown below.



Which of the following statements are true?

- 1 A buffer exists at point L.
- 2 The isoelectric point of alanine occurs at pH 6.

The major organic species present at point J and K are:

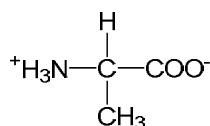
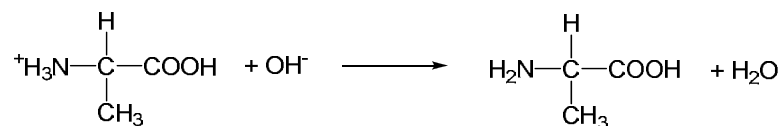


Ans: A

Thinking process: Identify the different species present at the various points of the titration curve. Fully-protonated alanine is in the conical flask, NaOH is in the burette.

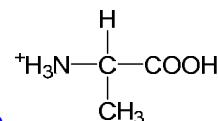
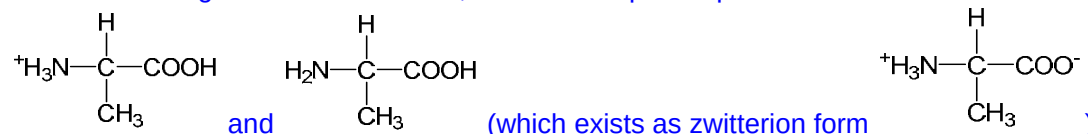
Hence, buffer only occurs AFTER the equivalence point. Hence option 1 is correct.

When fully protonated alanine reacts with NaOH, the following reaction occurs.



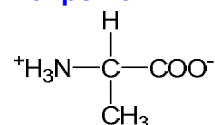
At the equivalence point, is present. Since this is a zwitterion, it is electrically neutral, hence the equivalence point pH can be taken to be the isoelectric point (the pH at which the amino acid is electrically neutral). Hence option 2 is correct.

Hence in the region from 0 to 25 cm³, there are 2 species present:



At point J: MAJOR organic species present is the protonated alanine.

At point K: The end point is nearing, hence the MAJOR organic species should be



Hence option 3 is correct.

Answers:

1	C	11	B	21	B	31	A
2	C	12	C	22	B	32	A
3	C	13	D	23	C	33	B
4	B	14	C	24	B	34	D
5	A	15	D	25	D	35	A
6	C	16	A	26	B	36	A
7	A	17	D	27	C	37	B
8	D	18	B	28	C	38	C
9	A	19	C	29	B	39	C
10	D	20	A	30	D	40	A